Model Question Paper – Cloud Data Engineer - Jain University

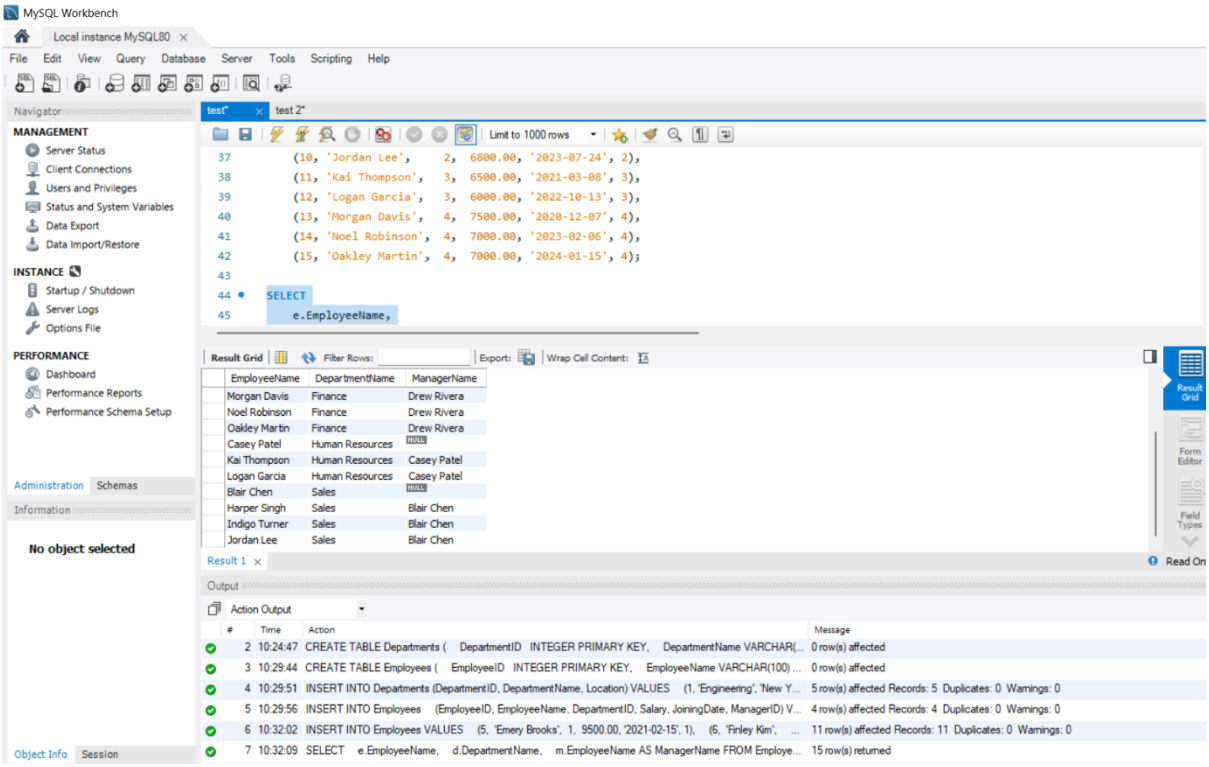
NAME : HARSHINI B

USN : 23BTRIS014

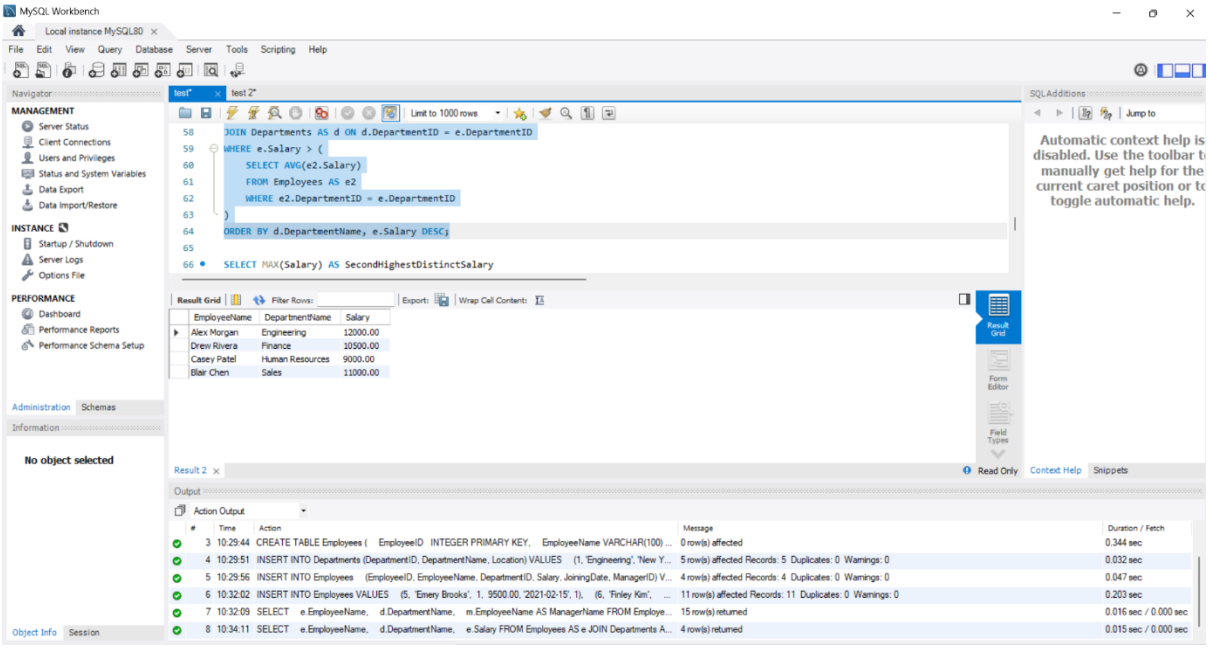
HCL ID : HCLTFP2260284

DEPT : ISE

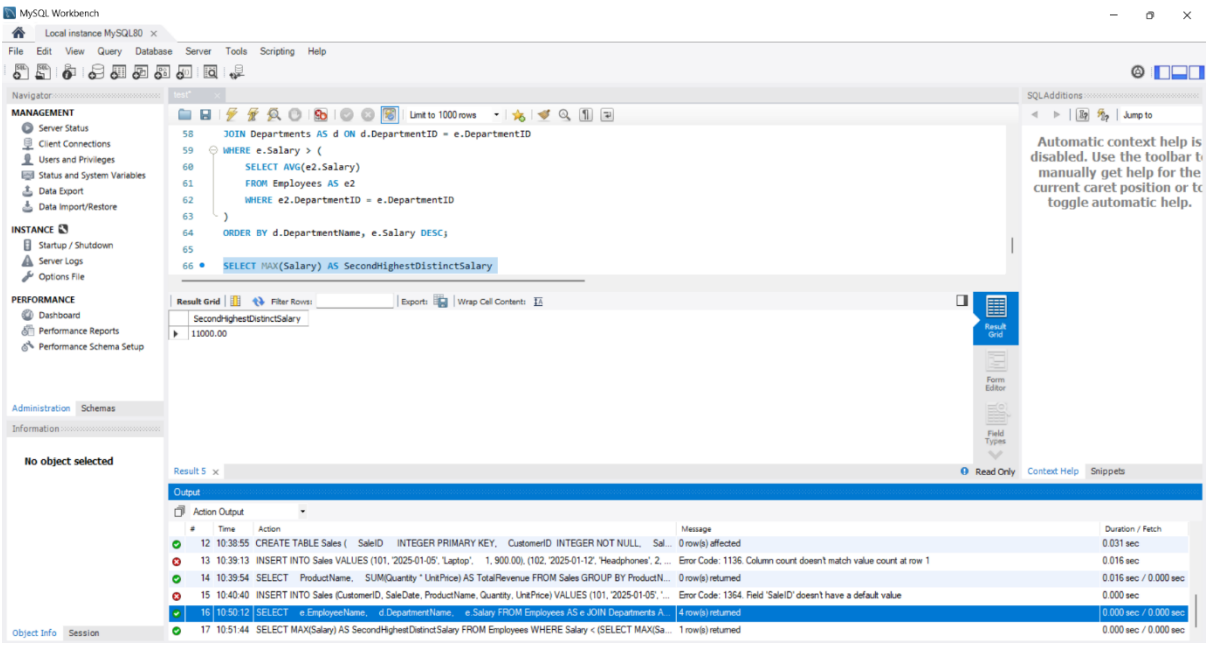
Tasks: a. Display employee names, department names, and manager names using joins. (2 marks)



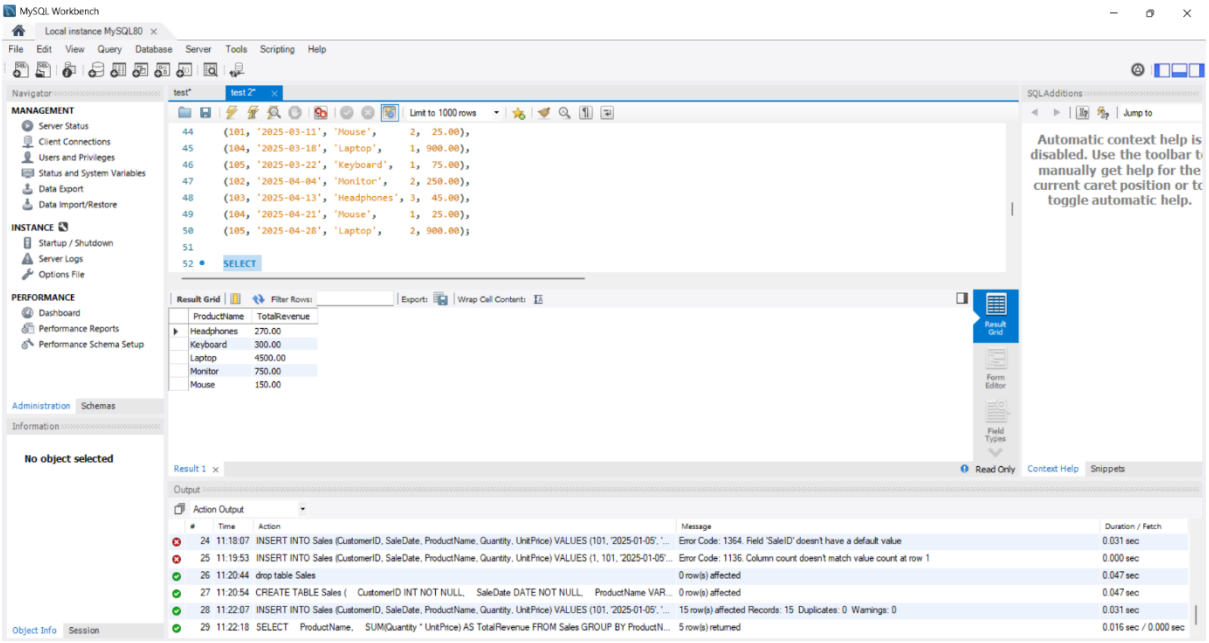
b. Find employees earning more than their department's average salary. (2 marks)



c. Find the second-highest distinct salary without using LIMIT, TOP, or FETCH FIRST. (2 marks)

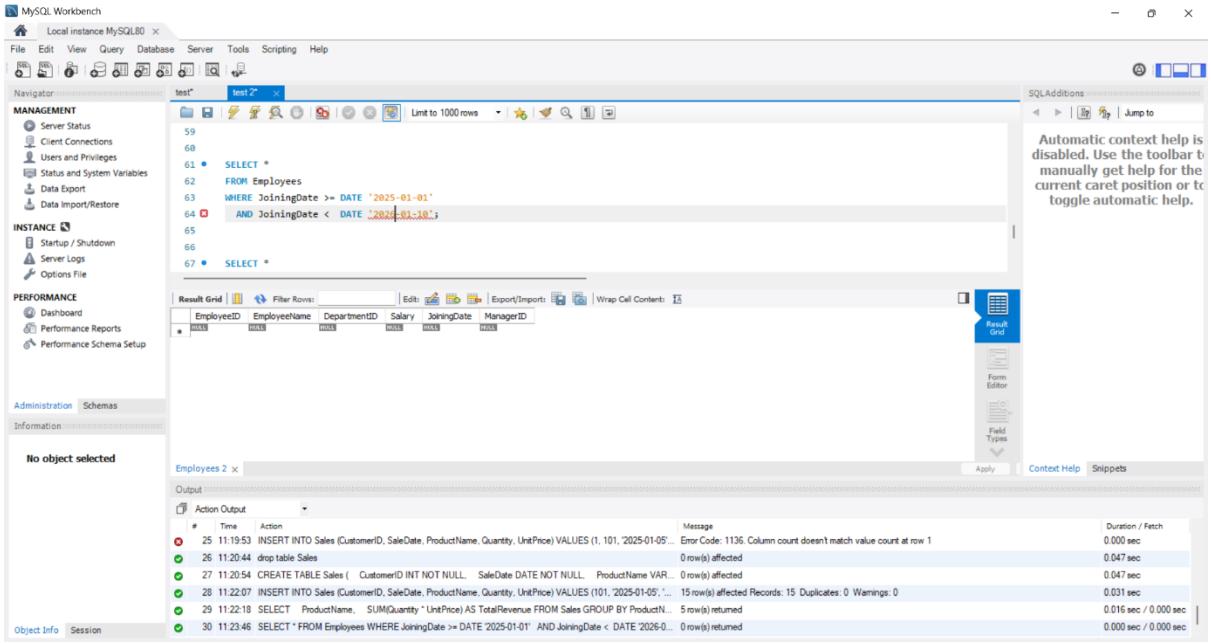


a. Calculate total revenue for each product.



b. Calculate monthly revenue and month-over-month revenue change using a window function

c. Rewrite the following query to improve the potential for index usage and explain why your version may perform better. (2 marks)
SELECT * FROM Employees WHERE YEAR(JoiningDate) = 2025;
Assume JoiningDate may be indexed.



Question 3: SQL Troubleshooting (6 Marks)

a. Explain how an execution plan helps identify expensive operations in a query. (2 marks)

A Sql plan helps us to identify how it is going to work . Using this execution plan we can identigy better query for better output.

c. Write a query to identify duplicate customer email addresses and display only addresses occurring more than once. (2 marks)

```
SELECT EmailAddress
FROM Customers
GROUP BY EmailAddress
HAVING COUNT(*) > 1;
```

SECTION B: LINUX || Maximum Marks: 15

a. Find all lines containing ERROR.

```
grep 'ERROR' /var/log/application.log
```

b. Display the 20 most recent error entries in the file. (1 mark)

grep 'ERROR' /var/log/application.log | tail -n 20

c. Count the total number of error entries. (1 mark)

grep -c 'ERROR' /var/log/application.log

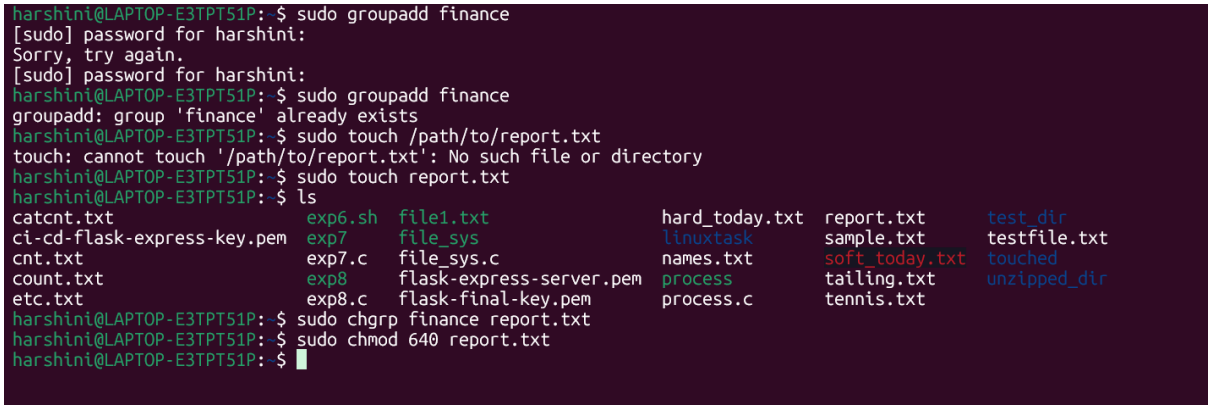
e. Explain how you would calculate the number of errors occurring in each hour. (1 mark)

I will count errors by date and hour for easy calculating

Question 5: Permissions and System Troubleshooting (5 Marks)

a. Create a group named finance and a report file, assuming you have the required privileges.

b. Set permissions so the owner can read and write, the finance group can read, and other users have no access. (1 mark)



c. Identify the processes consuming the most CPU.

```
harshini@LAPTOP-E3TPT51P:~$ free -h
df -h
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       416Mi       3.2Gi       3.5Mi       178Mi       3.3Gi
Swap:          1.0Gi          0B       1.0Gi
Filesystem      Size  Used Avail Use% Mounted on
none            1.9G     0  1.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            1.9G  4.0K  1.9G   1% /mnt/wsl
drivers         475G  426G   50G  90% /usr/lib/wsl/drivers
/dev/sdd       1007G  3.0G  953G   1% /
none            1.9G   80K  1.9G   1% /mnt/wslg
none            1.9G     0  1.9G   0% /usr/lib/wsl/lib
rootfs          1.9G  2.7M  1.9G   1% /init
none            1.9G  512K  1.9G   1% /run
none            1.9G     0  1.9G   0% /run/lock
none            1.9G     0  1.9G   0% /run/shm
none            1.9G   76K  1.9G   1% /mnt/wslg/versions.txt
none            1.9G   76K  1.9G   1% /mnt/wslg/doc
C:\             475G  426G   50G  90% /mnt/c
tmpfs           379M   20K  378M   1% /run/user/1000
harshini@LAPTOP-E3TPT51P:~$
```

e. Find files larger than 100 MB under a specified application directory.

find /path/to/application -type f -size +100M -print

Question 7: PySpark DataFrame Operations (6 Marks)

```
>>> df.orderBy(col("Salary").desc()).show(5, truncate=False)
+----------+------------+----------+-------+-----------+
|EmployeeID|EmployeeName|Department|Salary |JoiningDate|
+----------+------------+----------+-------+-----------+
|106       |Sneha       |IT        |90000.0|2019-11-25 |
|115       |Gokul       |Finance   |85000.0|2019-02-28 |
|103       |Karthik     |IT        |82000.0|2020-06-20 |
|113       |Prakash     |IT        |78000.0|2021-10-22 |
|101       |Arun        |IT        |75000.0|2022-01-15 |
+----------+------------+----------+-------+-----------+
only showing top 5 rows
>>>
>>> spark.stop()
```


```
>>> df.orderBy(col("Salary").desc()).show(5, truncate=False)
+----------+------------+----------+-------+-----------+
|EmployeeID|EmployeeName|Department|Salary |JoiningDate|
+----------+------------+----------+-------+-----------+
|106       |Sneha       |IT        |90000.0|2019-11-25 |
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>>> spark.stop()
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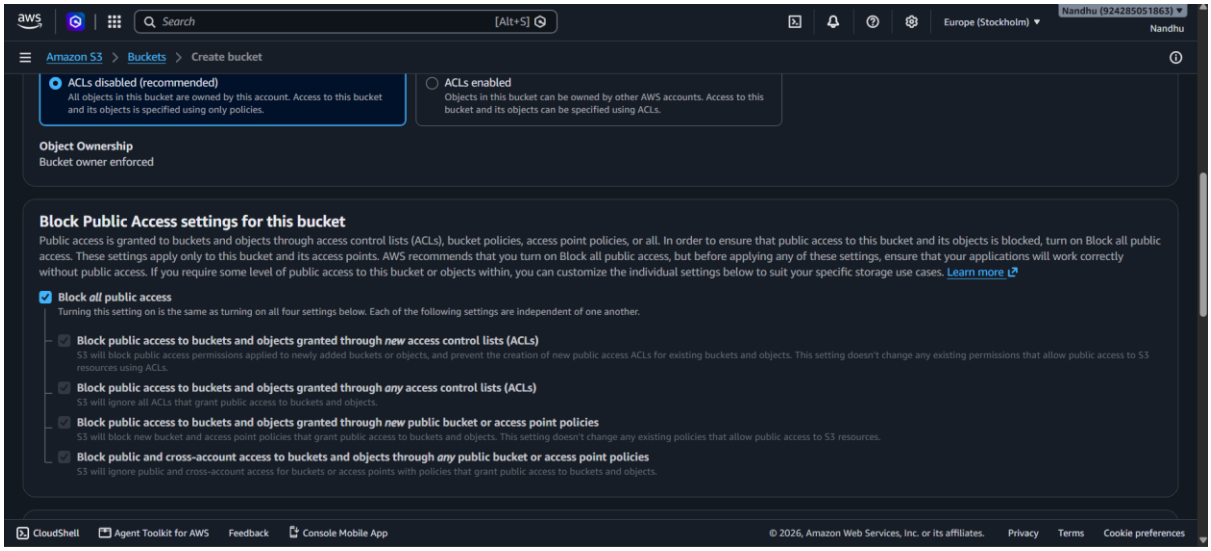
```
>>> df.groupBy("Department").agg(
...     round(avg("Salary"), 2).alias("AverageSalary"),
...     count("*").alias("EmployeeCount")
... ).orderBy("Department").show(truncate=False)
+----------+-------------+-------------+
|Department|AverageSalary|EmployeeCount|
+----------+-------------+-------------+
|Finance   |70000.0      |4            |
|HR        |52666.67     |3            |
|IT        |78600.0      |5            |
|Sales     |55333.33     |3            |
+----------+-------------+-------------+
```

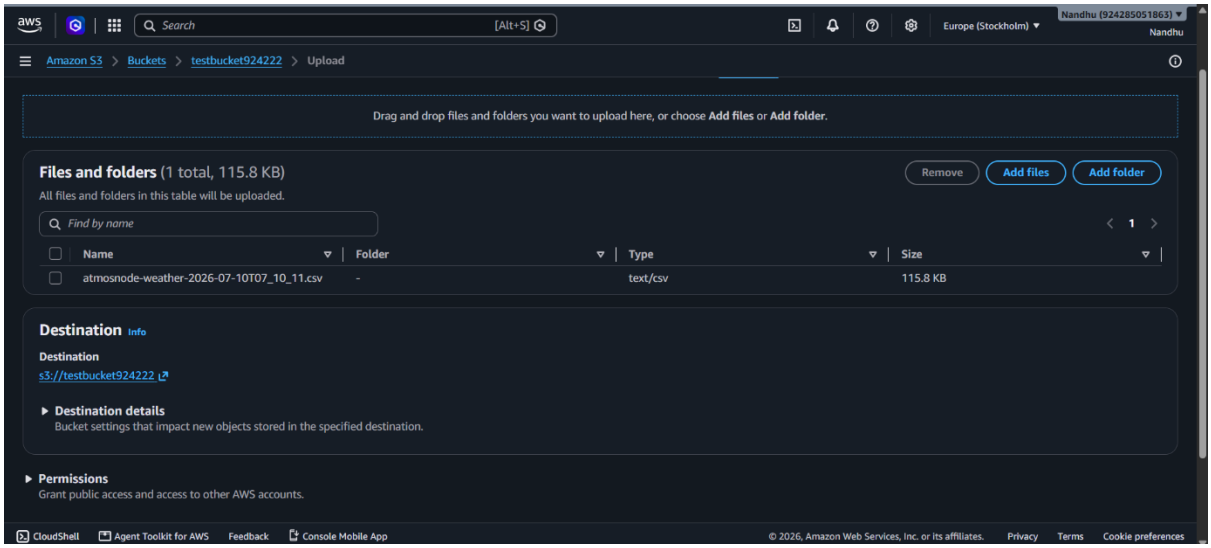
```
>>> df_annual.show(truncate=False)
+----------+------------+----------+------+-----------+------------+
|EmployeeID|EmployeeName|Department|Salary |JoiningDate|AnnualSalary|
+----------+------------+----------+------+-----------+------------+
|101       |Arun        |IT        |75000.0|2022-01-15 |900000.0    |
|102       |Priya       |HR        |45000.0|2021-03-10 |540000.0    |
|103       |Karthik     |IT        |82000.0|2020-06-20 |984000.0    |
|104       |Divya       |Finance   |65000.0|2023-02-05 |780000.0    |
|105       |Rahul       |HR        |52000.0|2022-07-12 |624000.0    |
|106       |Sneha       |IT        |90000.0|2019-11-25 |1080000.0   |
|107       |Vijay       |Finance   |58000.0|2021-08-18 |696000.0    |
|108       |Meena       |IT        |68000.0|2023-05-14 |816000.0    |
|109       |Suresh      |Sales     |48000.0|2020-09-01 |576000.0    |
|110       |Anitha      |Sales     |55000.0|2022-12-11 |660000.0    |
|111       |Manoj       |Finance   |72000.0|2018-04-16 |864000.0    |
|112       |Kavya       |HR        |61000.0|2024-01-08 |732000.0    |
|113       |Prakash     |IT        |78000.0|2021-10-22 |936000.0    |
|114       |Nithya      |Sales     |63000.0|2023-07-19 |756000.0    |
|115       |Gokul       |Finance   |85000.0|2019-02-28 |1020000.0   |
+----------+------------+----------+------+-----------+------------+
```

```
>>> df.filter(col("Salary") > 60000).show(truncate=False)
+----------+------------+----------+------+-----------+
|EmployeeID|EmployeeName|Department|Salary |JoiningDate|
+----------+------------+----------+------+-----------+
|101       |Arun        |IT        |75000.0|2022-01-15 |
|103       |Karthik     |IT        |82000.0|2020-06-20 |
|104       |Divya       |Finance   |65000.0|2023-02-05 |
|106       |Sneha       |IT        |90000.0|2019-11-25 |
|108       |Meena       |IT        |68000.0|2023-05-14 |
|111       |Manoj       |Finance   |72000.0|2018-04-16 |
|112       |Kavya       |HR        |61000.0|2024-01-08 |
|113       |Prakash     |IT        |78000.0|2021-10-22 |
|114       |Nithya      |Sales     |63000.0|2023-07-19 |
|115       |Gokul       |Finance   |85000.0|2019-02-28 |
+----------+------------+----------+------+-----------+
```

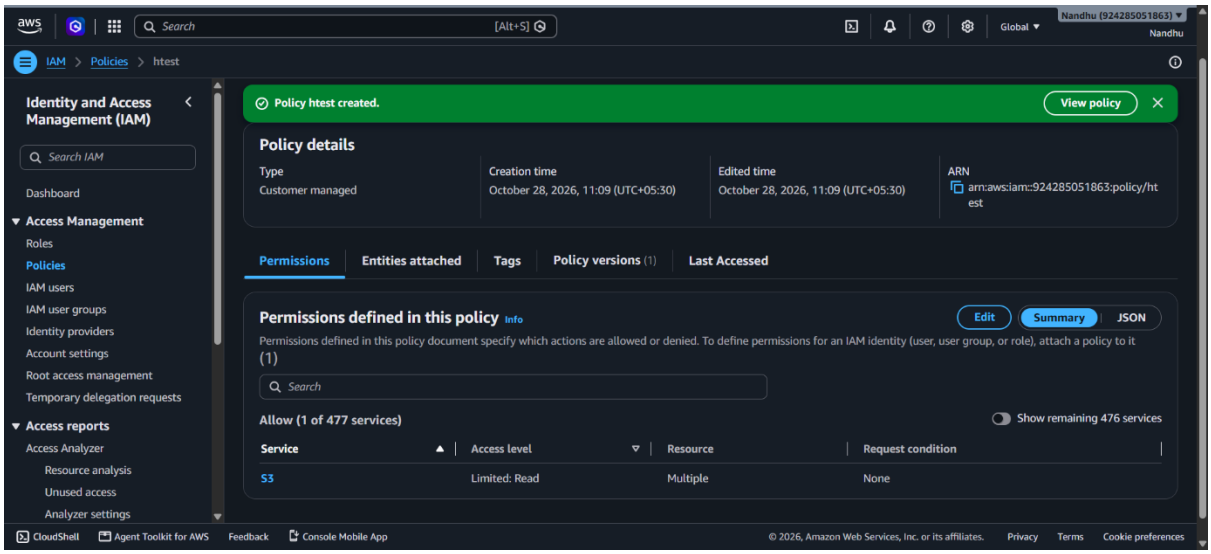
SECTION D: AWS CLOUD SERVICES

a. Create an S3 bucket with Block Public Access enabled and upload a sample file. (1 mark)





b. Design an IAM policy that grants read access to objects in the designated bucket without granting access to unrelated buckets. (2 marks)

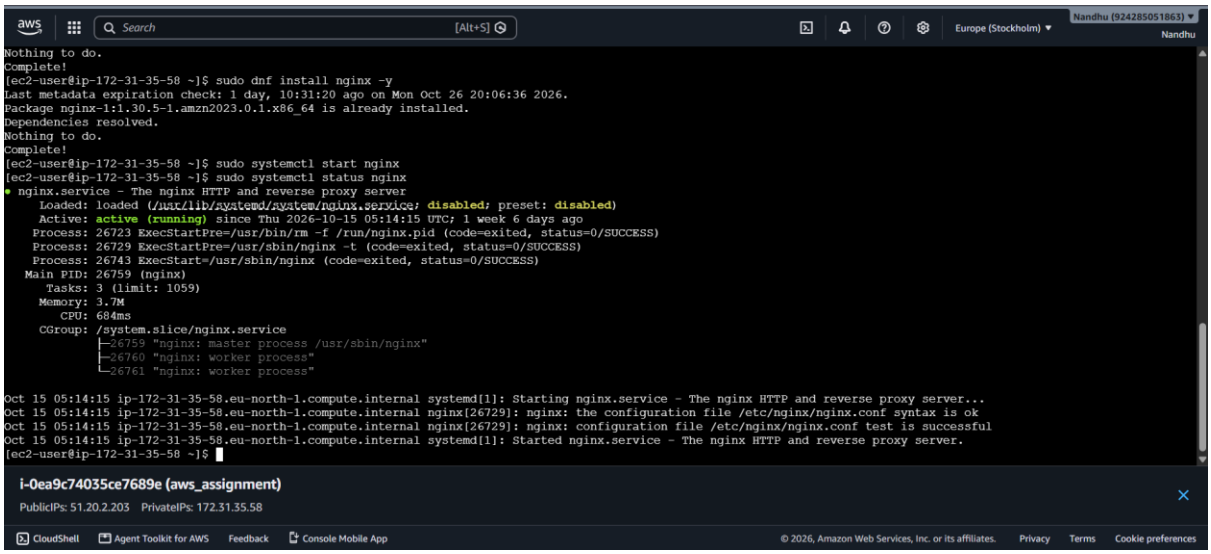


c. Explain how you would verify that an unauthorized user cannot read the file. (1 mark)

By using Access control List we can verify the unauthorized

Question 11: EC2 Troubleshooting and CloudWatch (5 Marks)

a. Check the EC2 instance state and status checks. (1 mark)



b. Explain how security groups and route tables can affect application connectivity. (1 mark)

Security Group acts as a firewall , it has inbound rules to connect to the network

Route table act as a network gateway , which helps the instance to connect without any traffic

c. Explain how to configure a CloudWatch alarm when CPU utilization exceeds 85% for a defined evaluation period. (2 marks)

1.Open the CloudWatch console

2.choose Alarms

3.click Create alarm

4.Select the AWS/EC2

5.choose the CPUUtilization metric for your specific instance and set the statistic to Average

d. Identify one method for investigating the root cause of high CPU utilization. (1 mark)

we can use aws system manager for investigating the root cause of high CPU Utilization

e. How encryption at rest protects stored data:

Encryption at rest protects stored data by converting the file contents into unreadable ciphertext.that makes

Question 12: Event-Driven Processing (5 Marks)

a. We can use AWS Lambda to automatically check and process CSV files without managing any servers.

b. Whenever a CSV file is uploaded to S3, it automatically triggers the Lambda function to start processing.

c. We need to give IAM permission to read files from S3 and save the processed results back to S3.

d. We can use CloudWatch to check the processing logs and find errors if something goes wrong.

e. We can track already processed files to avoid duplicates and store the results in a separate S3 bucket to prevent repeated triggering.

Question 13: State Machine Design (7 Marks)

a. Suitable states: We can use Task states to validate the file, transform the data, calculate total sales, store results in S3, and send a notification.

b. Choice state: The Choice state checks whether the file is valid. If valid, it continues processing; otherwise, it moves to the failure state.

c. Retry and Catch: Retry runs a failed task again if the error is temporary. Catch handles permanent errors and sends the workflow to a failure-handling state.

d. Invoking services: Step Functions can call Lambda functions or other supported AWS services to perform each processing step automatically.

e. Execution history: Execution history shows each state, its input and output, and any errors, helping us identify and fix failures.

Question 14: Step Functions Implementation (8 Marks)

a. Amazon States Language (ASL) is a JSON-based language used to define workflows in AWS Step Functions. It describes the tasks, sequence of execution, and error-handling methods**.**

b. Retry policy: The TransformData task retries temporary AWS service errors and timeouts up to three times, waiting two seconds initially and doubling the delay after each attempt.

c. Catch transition: If LoadResults fails, Catch sends the workflow to RecordFailure, which records the error before the workflow fails.

d. Passing data: Each task receives the previous task's output as its input by default. We can use ResultPath and OutputPath to control which data is passed.

e. Testing: Run one execution with valid input to check success, then test invalid input and simulated task failures to verify the failure, retry, and Catch paths.

Question 15: Azure Data Lake Storage and Security (5 Marks)

a. Create an Azure Storage Account and enable Hierarchical Namespace for ADLS Gen2.

b. Create a filesystem or container and upload a sample CSV dataset.

c. Microsoft Entra ID identifies users, and Azure RBAC assigns permissions to access the data.

d. Test access using an unauthorized account and verify that access is denied.

Question 16: Azure VM Troubleshooting (5 Marks)

a. Check the VM's provisioning state and power state in the Azure Portal.

b. Incorrect Network Security Group rules may block SSH or RDP connections.

c. Incorrect subnet settings or route tables can prevent the VM from connecting to the network.

d. Use Azure Bastion to connect securely without exposing the VM's public IP. Also check boot diagnostics and network settings.

Question 17: Azure Data Integration (5 Marks)

a. Use Azure Data Factory to automate and manage data movement.

b. Create a pipeline to copy the CSV file from its source into Azure Data Lake Storage Gen2.

c. Organize files into date-based folders, such as year/month/day.

d. Use mapping data flows to handle malformed records and replace or remove missing values.

e. Monitor pipeline runs in Azure Data Factory and validate the row count to ensure all records are copied.